(AP)(19.2)(27) 10.16. 2026

less actin filaments →

The mycardial conducting cells act almost → like neurons → propagated the action potential. (electric impulse) → travels throughout the contractions that propel the blood.

Structure of Cardiac Muscle → Compared to giant cylinders

of skeletal muscle, cardiac muscle cells, or cardiomyocytes → shorter with smaller diameters

Cardiac muscle also demonstrates striations the alternating pattern of (Dark → A) bands light

I-bands → attributed to the precise arrangem- of myofilaments & fibrils that are organized in sarcomeres along the length of the cell

★ Virtually identical to skeletal.

These contractile elements are virtually identical to skeletal muscle T-(traverse) ~~tubes~~ tubules → penetrate the surface plasma membrane the sarcolemma, to the interior of the cell, allowing the electrical impulse to reach the interior

The T-tubules are only found @ the Z-discs, whereas in skeletal muscle, they are found @ junction of A & I bands

There are ~~are~~ one-half as many T-tubules in cardiac muscles ~~in~~ as in ~~skeletal~~ muscle.

Additionaly → the sarcaplasmic reticulum stores few ~~calcium~~ calcium ions, so most of plentiful, providing energy for the contractions of the heart. Typically, cardiomyocytes have single (central nucleus) but some have two / more